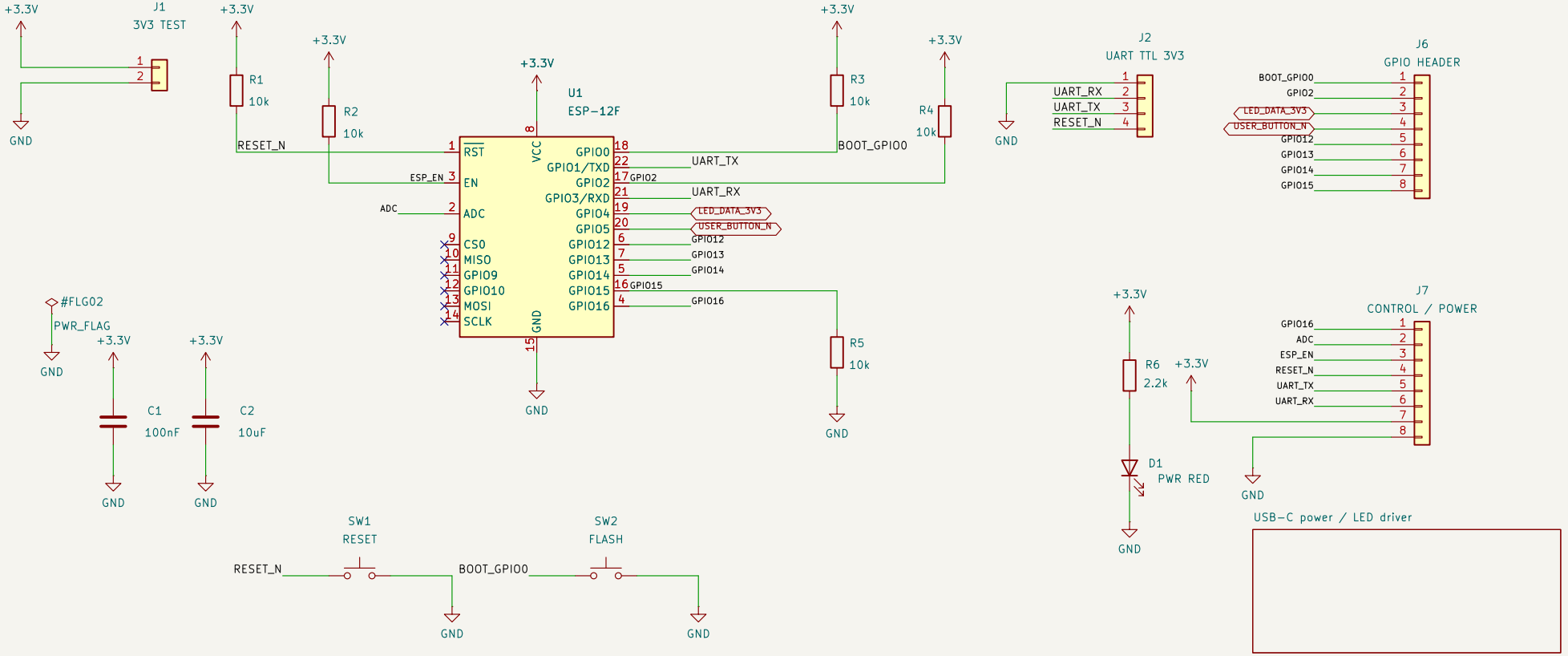
RGB Sleep Lamp — ESP-12F controller

J1: 3.3V test header. Supplied from power sheet; do not feed from UART adapter.



J2: pin 2 <- adapter TX; pin 3 -> adapter RX. Common GND.

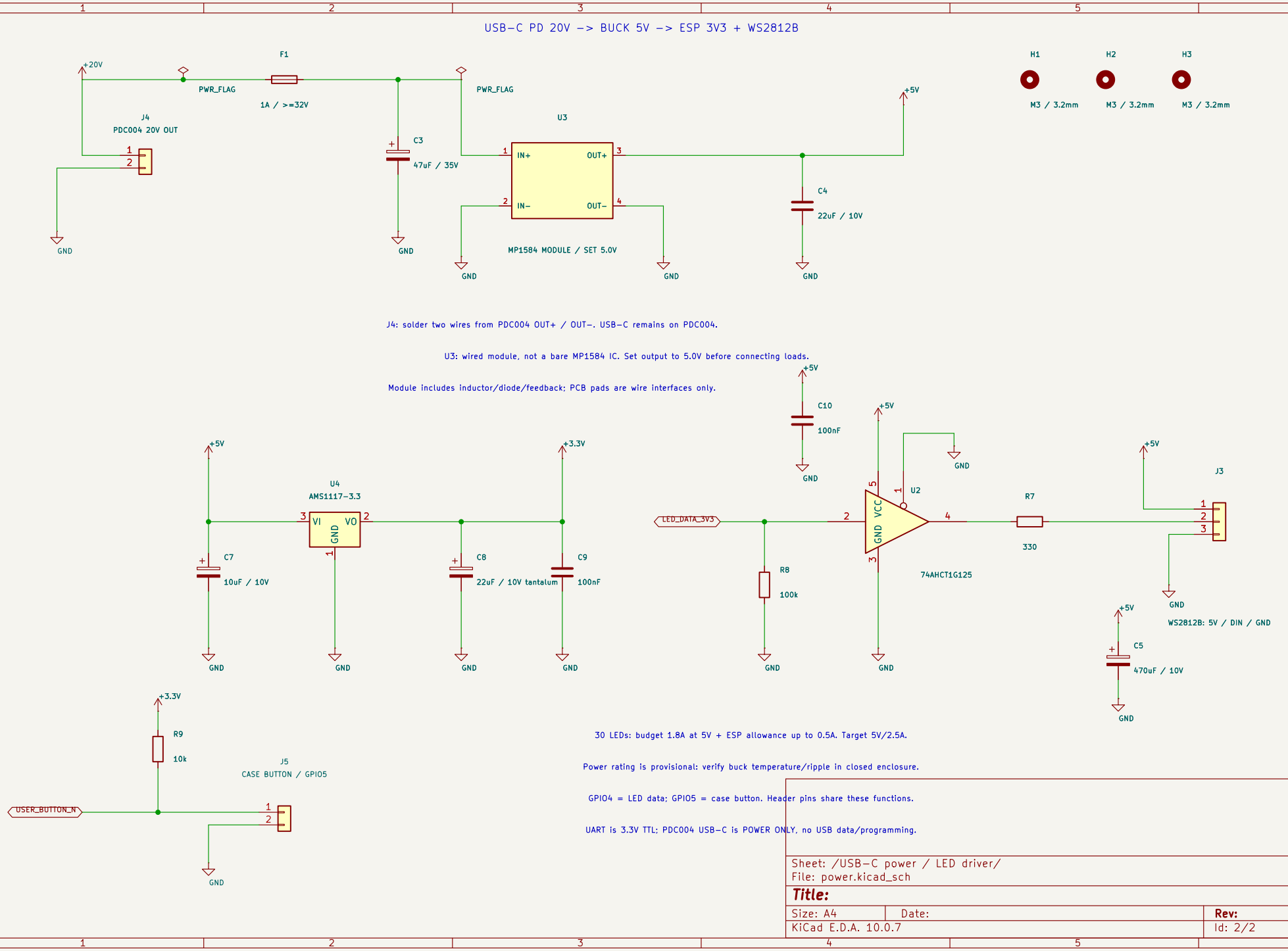
UART logic 3.3V only. Real RS-232 requires a level converter.

Upload: hold FLASH, pulse RESET, release FLASH, upload. Then RESET.

Place C1/C2 near module VCC/GND. ADC: 0-1V maximum at module pin.

PWR LED: red low-current type; ~0.6mA nominal (Vf=2.0V).

Sheet: / File: circuit.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 1/2


USB-C PD 20V -> BUCK 5V -> ESP 3V3 + WS2812B
+20V
F1
PWR_FLAG
1A / >=32V
PWR_FLAG
J4
PDC004 20V OUT
C3
47uF / 35V
U3
IN+
IN-
OUT+
OUT-
MP1584 MODULE / SET 5.0V
+5V
C4
22uF / 10V
GND
H1
H2
H3
M3 / 3.2mm
M3 / 3.2mm
M3 / 3.2mm
J4: solder two wires from PDC004 OUT+ / OUT-. USB-C remains on PDC004.
U3: wired module, not a bare MP1584 IC. Set output to 5.0V before connecting loads.
Module includes inductor/diode/feedback; PCB pads are wire interfaces only.
+3.3V
U4
AMS1117-3.3
VI
VO
GND
C7
10uF / 10V
C8
22uF / 10V tantalum
C9
100nF
C10
100nF
U2
VCC
GND
74AHCT1G125
LED_DATA_3V3
R7
330
R8
100k
J3
WS2812B: 5V / DIN / GND
C5
470uF / 10V
R9
10k
J5
CASE BUTTON / GPIO5
USER_BUTTON_N
30 LEDs: budget 1.8A at 5V + ESP allowance up to 0.5A. Target 5V/2.5A.
Power rating is provisional: verify buck temperature/ripple in closed enclosure.
GPIO4 = LED data; GPIO5 = case button. Header pins share these functions.
UART is 3.3V TTL; PDC004 USB-C is POWER ONLY, no USB data/programming.
Sheet: /USB-C power / LED driver/
File: power.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.7
Id: 2/2